Higher National Diploma in Quantity Surveying

First Year, Second Semester Examination - 2022

QS 12044: Civil Engineering Construction I

Instructions for Candidates:
Answer five questions only
All questions carry equal marks.

No. of questions: 06
No. of pages: 03
Time: 03 hours

1.

I. State the role of a Civil Engineer in Building Construction projects. (4 marks)

II. Mention types of information collected during the "Desk study" stage in site investigations. (5 marks)

III. Write purposes of soil investigation prior to start the building construction. (4 marks)

IV. Discuss the importance providing space for the Storage of Materials and equipment at the construction site. (3 marks)

V. Briefly explain general principles should be borne in mind in the selection of a site for building construction. (4 marks)

(Total 20 Marks)

2.

I. State the function of a building foundation. (3 marks)

II. Mention advantages of "Concrete" as a construction material in foundation. (3 marks)

III. State situations that require the construction of pile foundations? (4 marks)

IV. Safety in a construction site is essential for numerous reasons. Justify this statement by mentioning three common types of accidents and the relevant precautions in the construction of a multistory building. (6 marks)

V. Briefly discuss about the "Raft foundation". (4 marks)

(Total 20 Marks)

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3.

I. State the requirements of a wall. (4 marks)

II. State practices that should be considered in wall construction. (4 marks)

III. Briefly explain "Dampness" in a building. (4 marks)

IV. State effects of dampness on the building structure. (4 marks).

V. Well-designed Septic tank and soakage pit can be used to minimize the environmental pollution. Justify this statement. (You may use suitable sketches if needed) (4 marks)

(Total 20 Marks)

4.

I. State the stages involved in awarding a building construction project to a contractor. (6 marks)

II. Write responsibilities of following personnel in a construction site.

a) Consultant b) Project Manager

c) Quantity Surveyor d) Environmental Engineer (8 marks)

III. What is the difference between "Quality control" and "Quality assurance" in a building construction? Explain the importance of Quality assurance and Quality control. (6 marks)

(Total 20 Marks)

5.

I. State factors to be considered in selection of roof type (3 marks)

II. Discuss the purpose of installing a Ceiling for a building. (4 marks)

III. Briefly explain following types of roofs and their advantages.

a) Flat roofs b) Pitched roofs c) Shell roofs (9 marks)

IV. Discuss the purpose of scaffolding systems in building construction and state the primary types of scaffolding systems. (4 marks)

(Total 20 Marks)

6.

I. Briefly explain the utilization of the following construction machinery in building construction sites.

a) Excavator b) Crane

c) Porker Vibrator d) Ready mixed concrete truck (8 marks)

II. Write factors that should be considered in selecting suitable construction equipmen for the excavation work at the site. (4 marks)

III. Mention defects in floors of the buildings. (2 marks)

IV. Briefly explain the following work in floor construction.

a) Skirting Construction

b) Tiled floor construction

c) Cement concrete floor construction: (6 marks)

(Total 20 Mar